

Predictive Maintenance

End-to-End ML Lifecycle on Azure ML + MLflow

A complete MLOps lifecycle project: from dataset and experiments to registry, deployment, and monitoring.

Course
AIN-3009 Delivering AI Applications with
MLOps

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3.4%

failure rate

0.974

best CV ROC-AUC

v1

registered model

Succeeded

endpoint state

5-minute technical project presentation

Project Objective

What the assignment asks us to demonstrate

1

Experiment tracking

Log parameters, metrics, artifacts, and outputs with MLflow.

2

Model training and tuning

Train baselines and tune hyperparameters with traceable runs.

3

Model deployment

Package the model and serve predictions through an endpoint.

4

Performance monitoring

Track model behavior over time with drifted batches.

5

Model registry

Version the model and manage Staging → Production lifecycle.

Project framing

The rest of the deck maps each requirement to a concrete project artifact: code, metrics, MLflow logs, Azure registry, endpoint evidence, and monitoring charts.

Tracking

Training/Tuning

Registry

Deployment

Monitoring

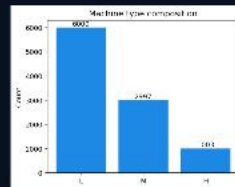
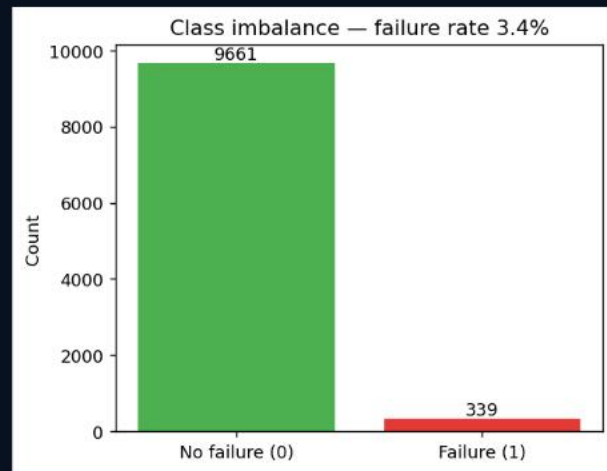
Dataset & Problem Framing

AI4I 2020 predictive-maintenance data: imbalanced binary classification

What data did we use?

- Source: AI4I 2020 Predictive Maintenance dataset
- 10,000 machine telemetry records
- Target: Machine failure (0/1)
- Failure rate is only 3.4%, so accuracy is misleading
- Leakage columns removed: TWF, HDF, PWF, OSF, RNF

Evaluation focus: ROC-AUC, PR-AUC, recall, and F1 instead of raw accuracy.



Tracking

Training/Tuning

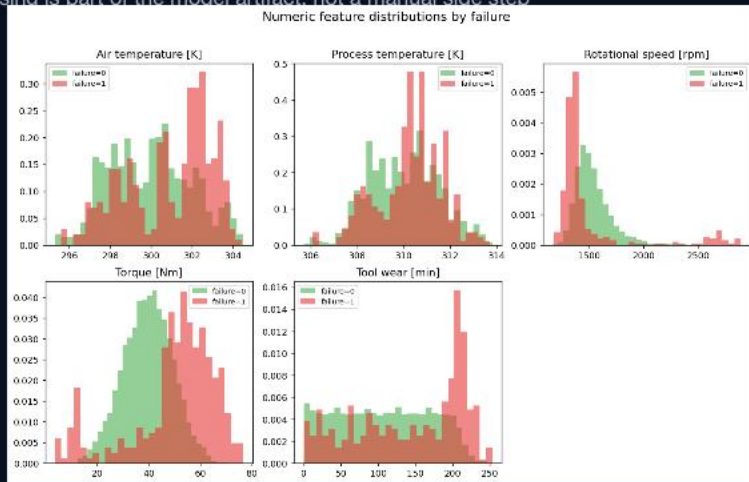
Registry

Deployment

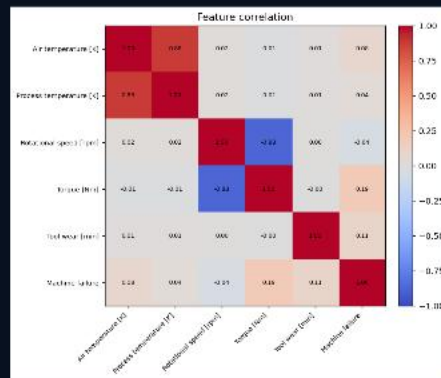
Monitoring

Data Preparation & EDA

Preprocessing is part of the model artifact, not a manual side step



The scikit-learn Pipeline serializes preprocessing with the estimator, so training and serving use the same transformations.

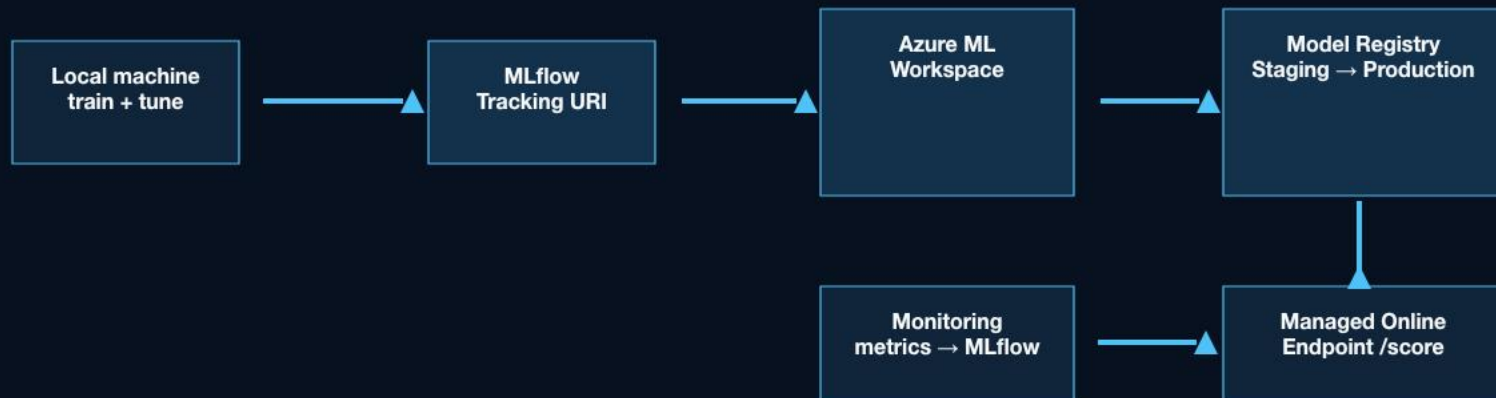


Preprocessing pipeline

- Numeric features → StandardScaler
- Machine Type → OneHotEncoder

Architecture: Local Training, Cloud Lifecycle

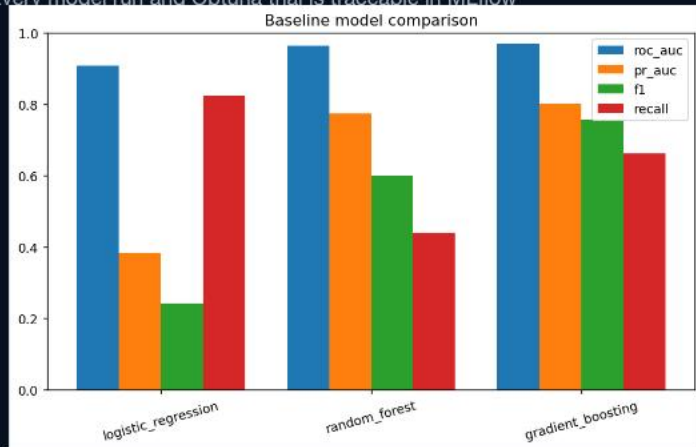
Azure credit is used for shared MLOps infrastructure, not expensive training compute



Stores: experiments, parameters, metrics, artifacts, registered models, and endpoint state.

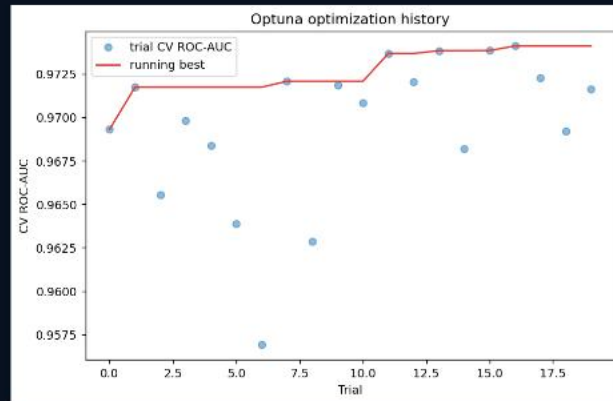
Training, Tracking & Tuning

Every model run and Optuna trial is traceable in ML flow



MLflow logs

- parameters, metrics, classification report, confusion matrix, serialized model artifact



20

Optuna trials

0.974

Best CV ROC-AUC

0.44 → 0.77

Recall lift

Tracking

Training/Tuning

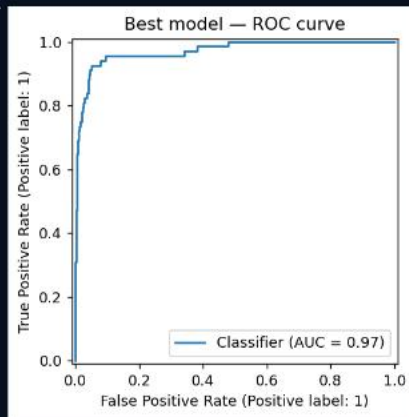
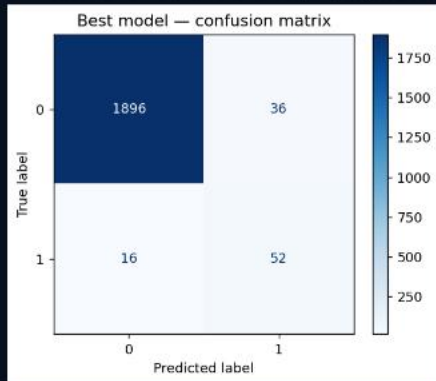
Registry

Deployment

Monitoring

Registry, Deployment & Pipeline Automation

The model becomes a versioned, deployable asset



PredictiveMaintenanceMode

Registered model

1

Version

Staging →
Production

Succeeded

Endpoint state

Airflow DAG orchestration

validate

train

tune

register

monitor

Tracking

Training/Tuning

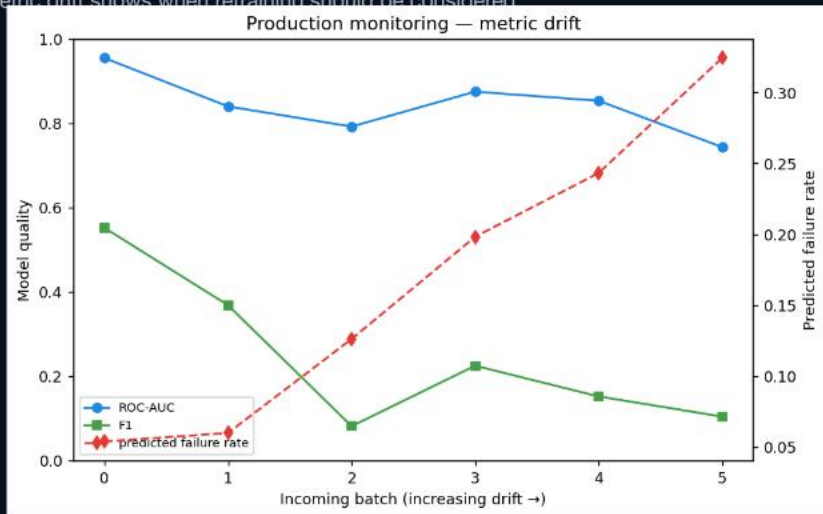
Registry

Deployment

Monitoring

Monitoring Closes the MLOps Loop

Metric drift shows when retraining should be considered



0.914

Batch 0 ROC-AUC

0.766

Batch 5 ROC-AUC

4% → 37%

Predicted failure rate

Course objectives covered

- ✓ Experiment tracking
- ✓ Training and tuning
- ✓ Registry and deployment
- ✓ Monitoring and drift tracking

Final takeaway: this is not a notebook-only model; it is an end-to-end MLflow + Azure ML lifecycle demonstration.

Tracking

Training/Tuning

Registry

Deployment

Monitoring